Assessment 1 - CS6163 Advanced Topics Seminars and Project Specification

Gabriel Espinoza Rojas, 22176691

0.1 Literature review

0.1.1 State of the art

EMAs

Ecological momentary assessments (EMAs) are a well-studied method that has proven useful in understanding an individual's state of mind (Künzler et al. 2019). EMAs are highly time-sensitive and considered a form of active digital phenotyping because they depend on user input. This can work against researchers, as individuals can grow tired or get bored of them.

King et al. acknowledges at the end of their research that the study carried out with 10 daily EMAs (13 questions) is unlikely to be successful with a similar frequency with complex questionnaires (King et al. 2024). However, active digital phenotyping is only a part of the story.

Passive digital phenotyping

Nowadays, it is not unusual to combine active and passive digital phenotyping, the latter being information collected from smartphones or wearable devices. In addition to EMA data, the receptivity study by Künzler et al. (2019) used **contextual factors** such as the person's location (GPS), their step count, date and time (in relation to notifications), smartphone's battery status and device usage metrics depending on OS (Wi-Fi connection state and Lock/Unlock events for iOS) (Künzler et al. 2019).

For post-study research however (Mishra et al. 2021), the passive features were narrowed by using only Apple phones and dropping the GPS data. While it is understandable to simplify the scope of the passive sensing data by discarding Android users, the justification to exclude the location data appears circumstantial.

A different study relied more on wearable devices (King et al. 2024): with a chest pad, a wristband and a research phone (second to their personal smartphone), King et al. gathered accelerometer, electrocardiography, and skin temperature data.

Interventions

Machine learning studies involving EMAs often also involve some type of Ecological momentary interventions (EMIs). Methods recently used in the domain of *mHealth* are often referred to as just-in-time interventions (JITI) or just-in-time adaptive interventions (JITAI):

> Failing to collect EMA responses may impede researchers' ability to identify real-world measures of health behaviors, and without participants receiving or engaging in JITIs, researchers may find it challenging to measure the efficacy of the intervention. (King et al. 2024, p. 2)

Quantifying the *receptivity* of interventions is a relatively new approach in *mHealth* but receptivity metrics often involve other concepts already extensively studied in a broader general approach to mobile notifications (Künzler et al. 2019).

Predicting receptivity

The researchers Künzler et al., Mishra et al. and King et al. all used *receptivity* as the predictor's target (*was the JITAI/EMA answered within 10 minutes?*). This provides an opportunity to partially compare the models, with the caveat that their respective training datasets differ in size (189, 141 and 45 participants) and contextual factors (personal iOS smartphones versus dedicated wearables).

The study *Ally+* (Mishra et al. 2021) reports a that the Random Forest (RF) model and a Support Vector Machine (SVM) model predicted the receptivity (metric `just in time respond`) with an F_1-score of 0.33 and 0.36, respectively. The score of the SVM matches the RF F_1-score (0.36) of the initial *Ally* study (Künzler et al. 2019). The subtleties between the use cases can explain this. RF in Künzler et al.'s work was a retrospective model. But the models in Mishra et al.'s follow-up work were pre-trained models prospectively deployed. This means that SVM and RF, despite being static, were part of the response workflow actively sending JITAIs.

Furthermore, the results reported by (Mishra et al. 2021) also include a Logistic Regression (LR) adaptive models `P1` and `P1+P2`, with the latter (0.38) beating all other previous F_1-scores. This is remarkable considering the model only learned approximately 50 sample points over the pre-training data ($\approx$ 600 points). Important to remark as well that technical problems substantially reduced the number of interventions triggered by these models (almost 50% in comparison with the static models).

Going back to retrospective modelling, (King et al. 2024) reports an RF and a Neural Network model (with 3 layers) predicting the receptivity (in this case not to a JITAI but to a following EMA) with an F_1-score of 0.87 and 0.86, respectively. The authors of this work note that no direct comparisons can be made across the models from (Mishra et al. 2021) due to the difference in granularity between datasets.

Although it is understandable that Künzler et al. was focused more on feasibility than cutting-edge technology, even the follow-up work by Mishra et al. and the separete study by King et al. selected machine learning models far from the state of the art in the realm of time series classification. This space is now dominated by deep learning solutions like CNNs and transformers (Wang et al. 2025). A potential next step for receptivity research in *mHealth* using ML is updating the models used so far.

References

King, Z. D., Yu, H., Vaessen, T., Myin-Germeys, I. and Sano, A. (2024), 'Investigating receptivity and affect using machine learning: Ecological momentary assessment and wearable sensing study', *JMIR mHealth and uHealth* **12**, e46347–. ii, iii

Künzler, F., Mishra, V., Kramer, J.-N., Kotz, D., Fleisch, E. and Kowatsch, T. (2019), 'Exploring the state-of-receptivity for mhealth interventions', *Proceedings of ACM on interactive, mobile, wearable and ubiquitous technologies* **3**(4), 1–27. ii, iii

Mishra, V., Künzler, F., Kramer, J.-N., Fleisch, E., Kowatsch, T. and Kotz, D. (2021), 'Detecting receptivity for mhealth interventions in the natural environment', *Proceedings of ACM on interactive, mobile, wearable and ubiquitous technologies* **5**(2), 1–24. ii, iii

Wang, Y., Wu, H., Dong, J., Liu, Y., Wang, C., Long, M. and Wang, J. (2025), 'Deep time series models: A comprehensive survey and benchmark'. iii